

Harmoniq

Software Version 1.1

The Ultimate Music App for Tabs and Sheet Music



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1. Introduction

Purpose

Harmoniq aims to provide a comprehensive platform for all musicians of all levels to access, create, and share tabs and sheet music. The app will help users improve their musical skills and simplify the process of learning or composing music. By combining essential features like editing tools and a library of resources, the app fills a gap in the market where current solutions often fall short.

The app's goal is to serve as a tool for students, professionals, and hobbyists, by making music practice and composition more accessible and engaging. Whether users are learning a new song, arranging music, or teaching students, this app will address their needs efficiently.

Scope

This document will cover:

- Stakeholders involved in the development and use of the app.
- Any constraints impacting the project.
- A competitive analysis of similar applications to outline our approach.

2. Stakeholders

Musicians:

- **Students:** Beginner and intermediate-level users learning new instruments or songs.
- **Professionals:** Composers, performers, and teachers requiring advanced tools for composing or instructing.
- **Hobbyists:** Individuals playing music for fun who need simple, accessible features.

Developers:

- The team responsible for designing, building, and maintaining the app.

Clients:

- The individuals or organizations funding the project or requesting the product.

Personas

Persona 1



Age

24 YEARS OLD

Name

Lewis Clark

Occupation

TEACHER

Location

Cincinnati, OH

Status

SINGLE

PERSONAS BIO

Lewis Clark is a high school music teacher looking for a modern way to keep his students interested in music. Lewis thinks that students would be more likely to stay focused with real-time feedback from an app instead of static sheet music on paper.

GOALS

- Share music easily with his class
- Give feedback to students
- Recording feature for students to submit playing assignments

MOTIVATIONS

- | | |
|------------------|-------------|
| • Comfort | <div></div> |
| • Convenience | <div></div> |
| • Price | <div></div> |
| • Tech expertise | <div></div> |

FRUSTRATIONS

- Paywalls
- Hard to use interfaces
- App Crashing/Downtime

PERSONALITY

- Smart
- Caring
- Musical

INTERESTS

- Performing
- Teaching
- Composing

Persona 2



Age

29 YEARS OLD

Name

Becci Anderson

Occupation

LAWYER

Location

Providence, RI

Status

MARRIED

PERSONAS BIO

Becci wants to balance her demanding career with her passion for music, which helps her unwind after long hours in the office. She relies heavily on technology to stay connected, organized, and entertained, and she's always on the lookout for tools that complement her lifestyle.

GOALS

- Discover new music
- help make a playlists for every mood
- learn something new

MOTIVATIONS

- | | |
|------------------|-------------|
| • Comfort | <div></div> |
| • Convenience | <div></div> |
| • Price | <div></div> |
| • Tech expertise | <div></div> |

FRUSTRATIONS

- Slow apps
- Annoyed by ads in apps
- Overwhelmed by choices

PERSONALITY

- Analytical
- Time-conscious
- driven

INTERESTS

- Legal podcasts and self-growth
- jazz instruments
- reading novels

Persona 3



Age

34 YEARS OLD

Name

Dale Whitfield

Occupation

MUSICIAN

Location

Nashville, TN

Status

MARRIED

PERSONAS BIO

Dale has been performing for over a decade and plays at local venues with his band. He often needs quick access to tabs or sheet music while on the road. Marcus values efficiency and quality in any music-related tool he uses.

GOALS

- Play Along with Tracks
- Streamline setlists
- Expand his Repertoire

MOTIVATIONS

- | | |
|------------------|-------------|
| • Comfort | <div></div> |
| • Convenience | <div></div> |
| • Price | <div></div> |
| • Tech expertise | <div></div> |

FRUSTRATIONS

- Ads or paywalls
- Lack of offline access
- Poorly transcribed tabs

PERSONALITY

- Detail oriented
- Slightly impatient when technology doesn't meet expectations

INTERESTS

- Classic rock
- Blues
- Gear reviews
- Live music events

3. Constraints

Time Constraints

- The project must be completed within the semester, with specific milestones due at regular intervals as specified in the course schedule.
- A final working prototype of the app is expected by the end of the course.

Monetary Constraints

- No monetary budget is available for the project.

Technical Constraints

- All code must be well-documented and follow standard coding conventions for Java.
- The app must be developed in Java, using tools such as VS code, Eclipse, or any other Java IDE approved by the course.

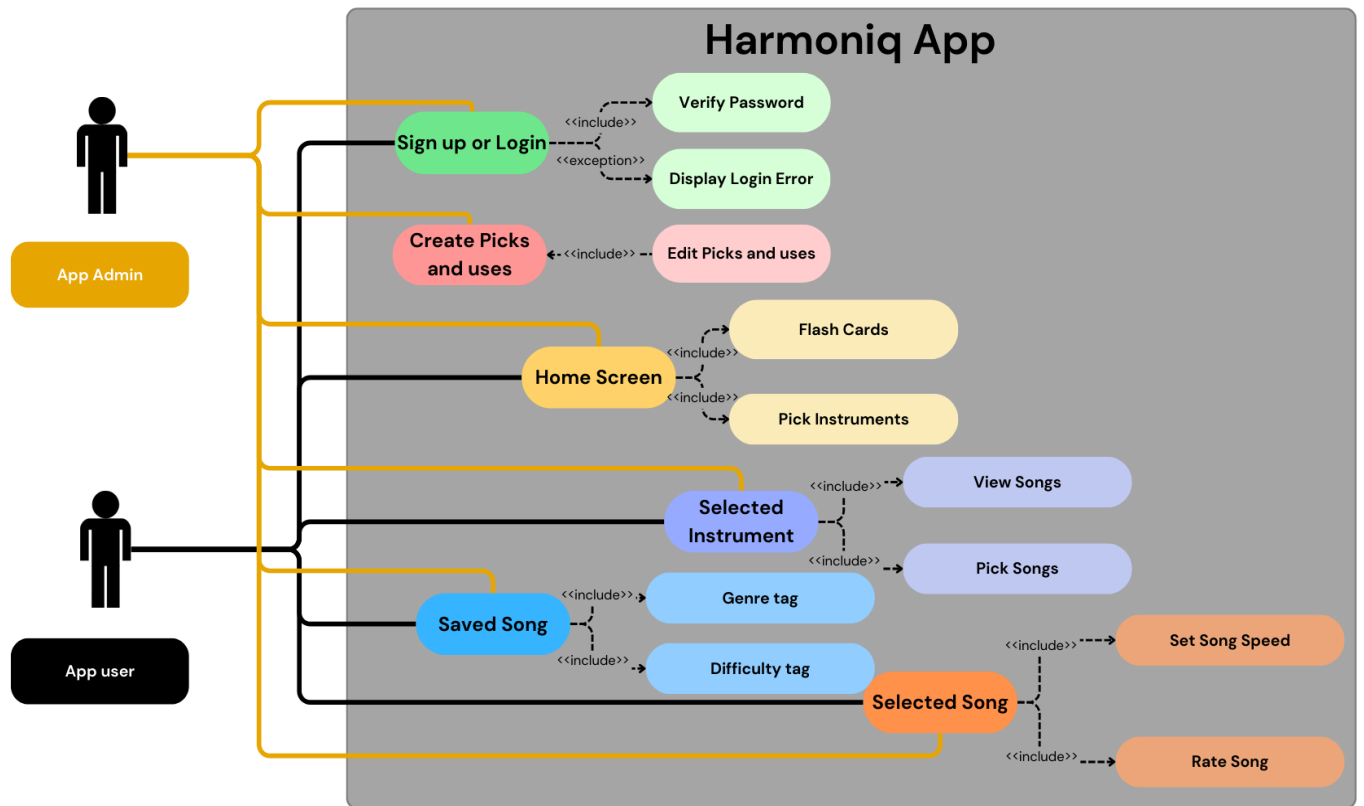
4. Overall Description

This app is designed to be accessed by users of all ages who are seeking to learn how to play music, regardless of their level of experience. The app will provide a platform for users to engage with musical learning materials, such as tabs and sheet music. The goal is to offer users the ability to explore a variety of resources that can support their musical growth and development.

The app will aim to allow users to access these materials conveniently and, where possible, provide ways for them to interact with or personalize their experience. Additionally, the app seeks to create a space that could enable users to share their musical content or engage with a broader community of learners.

While the exact features and capabilities of the app are still being determined, the primary focus is to create a tool that can serve as a valuable resource for musicians at various stages of their journey.

5. Business Use Cases



6. Functional Requirements

Requirements Spreadsheet

7. Non Functional Requirements

Look and Feel Requirements

- The UI should be modern and visually appealing.
- The app should have a simple and intuitive layout, avoiding unnecessary clutter.
- A consistent color scheme and typography should be used throughout the application.

Usability Requirements

- The app should be easy to navigate, with intuitive menus and clear visual indicators.
- Mobile-first responsive design should be implemented to ensure usability on smartphones and desktops.

Reliability Requirements

- User data should be saved properly and not lost when the app is refreshed or restarted.

Maintainability and Scalability Requirements

- The code should be well-organized and commented to allow future modifications.
- The app should be built using a simple, modular structure, making it easy to add new features if needed.
- The database and backend should handle at least a small number of simultaneous users without performance issues.

8. Definitions & Acronyms

Term	Definition
Tabs (Tablature)	Musical notation that uses instrument fingering rather than musical pitches. (Ex. Guitar)
Sheet Music	Traditional written notation of music that utilizes staff lines, notes, and rhythms.
DAW (Digital Audio Workstation)	A software application used to record, edit, and produce audio files.
MIDI (Musical Instrument Digital Interface)	A technical standard for computers to communicate musical information.
IDE (Integrated Development Environment)	A software containing tools for software development and code editing.
UI (User Interface)	The visual layout and interactive elements of an application.
Database	A structured set of data held in a computer or server. Harmoniq's database would store user information, song files, rating data, etc.
MusicXML	An open, XML-based file format for sharing and distributing musical scores.

9. Competitive Analysis

	Sibelius
Strengths	- Sibelius has long been considered an industry standard for music notation, making it widely trusted and accepted by professionals. - It offers an extensive library of tools and features for creating complex and professional-grade sheet music, including detailed notation for orchestras, bands, and choirs. - The playback feature allows users to hear their compositions with realistic instrument sounds, helping them refine their work. - It integrates well with other professional tools, such as digital audio workstations (DAWs) and MIDI controllers, enhancing its usability in studio and live settings. - Frequent updates ensure compatibility with modern operating systems and introduce new features.
Weaknesses	- The cost of Sibelius is one of its biggest downsides, with high upfront and subscription prices that may deter students, hobbyists, and small organizations. - Its interface can feel overwhelming to new users, especially those unfamiliar with music software. Tutorials and a learning period are often necessary to master it. - The software requires a relatively powerful computer to run smoothly, particularly when working with large, detailed scores.
Audience/Focus	- Sibelius is ideal for professional composers, arrangers, and music publishers who need advanced tools for creating and distributing sheet music. - It's also used by advanced music students and educators in academic settings due to its features for teaching and learning.

	Dorico
Strengths	- Dorico is designed with modern workflows in mind, making it more intuitive and easier to use compared to older notation software. Its clean and organized layout minimizes clutter. - The engraving capabilities are exceptional, producing beautifully formatted sheet music that requires little manual adjustment. This is especially helpful for publishing. - It includes unique features such as a focus on musical flow, enabling composers to work on sections of music seamlessly without worrying about page layout. - Dorico is highly customizable, allowing users to adapt the software to their preferred workflow.
Weaknesses	- As a relatively new product in the market, Dorico has a smaller user base, which means fewer community-created resources such as tutorials and forums. - Although it has a user-friendly interface, some advanced features (like custom notations) can be challenging for beginners to navigate. - It is priced at a premium level, which may make it less accessible to students or casual users who don't need advanced engraving capabilities. - While Dorico has been improving rapidly, some users may find certain features missing compared to more established competitors like Sibelius.
Audience/Focus	- Dorico is aimed at professional composers, engravers, and publishers who value precision and presentation. - It's an excellent choice for educators and students focused on producing high-quality sheet music for academic or performance purposes. - The software is also appealing to users who appreciate modern interfaces and workflows, particularly those who feel older software is outdated or clunky.

	MuseScore
Strengths	- MuseScore is completely free and open-source, making it accessible to anyone regardless of their budget. It's an excellent option for schools, students, and beginners who are just starting out with music notation. - The software has a very active and supportive online community that shares resources, such as public domain scores, tutorials, and advice for troubleshooting. - Despite being free, MuseScore offers a wide range of features, including playback, transposition, and support for various file formats like MIDI and MusicXML. - The interface is straightforward and user-friendly, making it easy for beginners to learn while still offering enough depth for more experienced users.
Weaknesses	- The software can struggle with performance when handling very large scores, occasionally leading to crashes or slowdowns. - There is no dedicated customer support, as users rely on community forums and documentation for help. This can be frustrating for those who encounter technical issues. - The playback feature, while useful, is less sophisticated compared to paid alternatives, as it doesn't use high-quality virtual instrument libraries.
Audience/Focus	- MuseScore is perfect for beginners, students, and casual musicians who need a free and reliable tool for creating and sharing sheet music. - It's also a good option for hobbyists who don't require the advanced features of paid software. - Schools and community music programs often use MuseScore due to its cost-effectiveness and ease of use.

Summary

	Strengths	Weaknesses	Audience/Focus
Sibelius	Industry-standard, advanced features, high-quality playback, and versatile tools.	Expensive, steep learning curve, and is hardware-dependent.	Professionals, advanced students, music publishers, and educators.
Dorico	Modern design, polished sheet music, real-time MIDI input, and workflow-friendly.	Smaller user base, fewer tutorials, is pricey, and some features are still in progress.	Professionals, publishers, educators, and users who prefer streamlined modern tools.
MuseScore	Free, user-friendly, active community, suitable for beginners and schools.	Limited advanced features, slower with large files, and lacks direct support.	Beginners, students, hobbyists, schools, and those looking for a budget option.

10. References

Competitive Analysis Sources:

- [Sibelius](#)
- [Dorico](#)
- [Musescore](#)
- [SoundOnSound](#)

11. Appendices

Appendix A: Instrument Support

- The application may support a range of musical instruments for notation and playback. The final selection of instruments is still being determined based on project scope and feasibility.

Appendix B: User Roles

- Standard Users: Can browse and edit music tabs.
- Administrators: Have the ability to manage content and moderate user submissions.